

IIT Hyderabad Innovation Ecosystem

Inventing & Innovating in Technology for Humanity



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad



Dean Innovation Office

IITH Innovation in Numbers

Incubators	04	Patents Filed	610+
Startups	300+	Technology Transfers	09+
Patents Granted	230+	Revenue Generated by startups	2000 Cr+

Data as of FY 2024-25

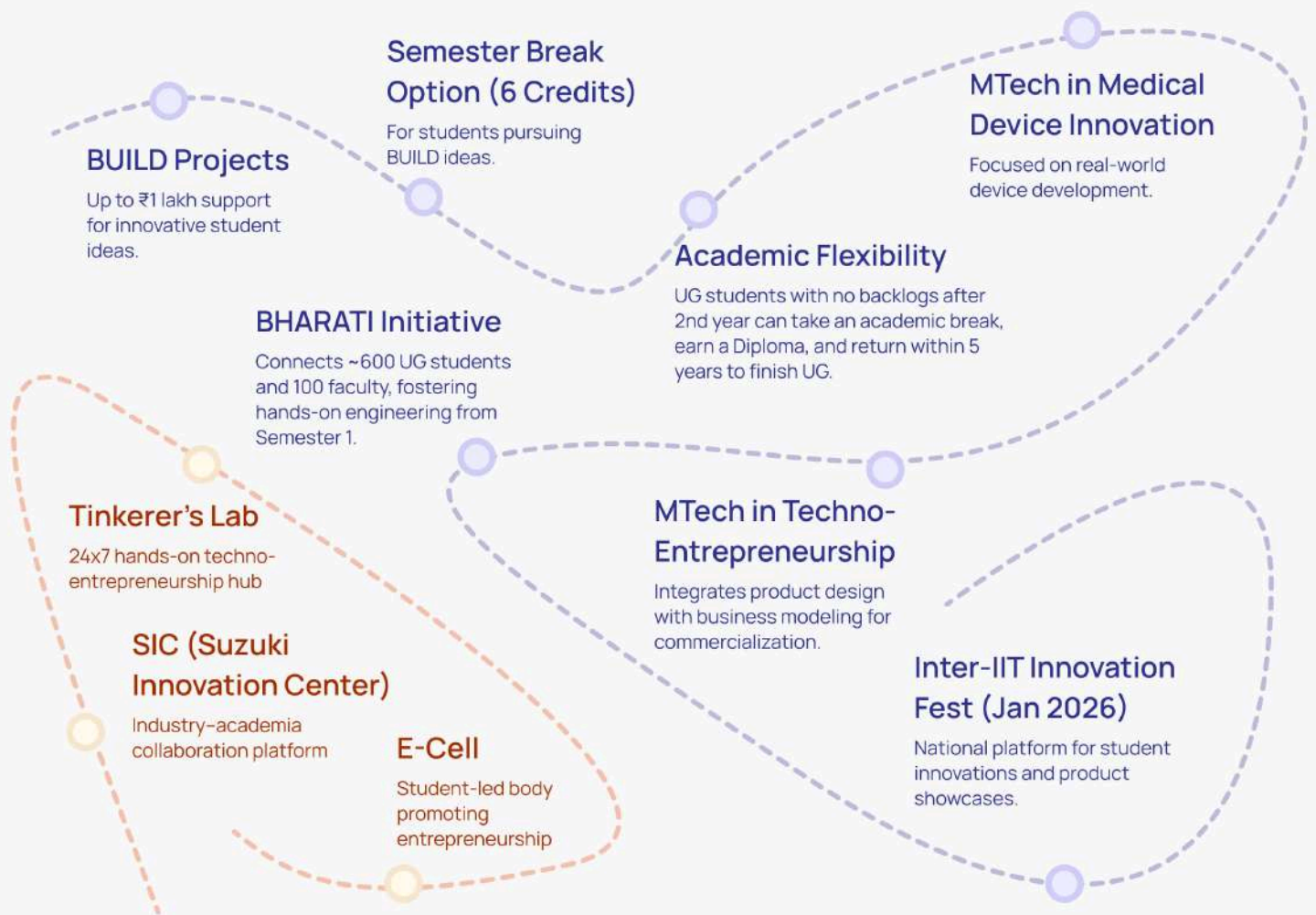
NICE Program (NMDC & iTIC Incubator)

Total funds disbursed	4.9 Cr	Startups supported	17
Revenue Generated by startups	8.0 Cr	Jobs Created	~120
Interns Hired	~ 100	Valuation	~75 Cr

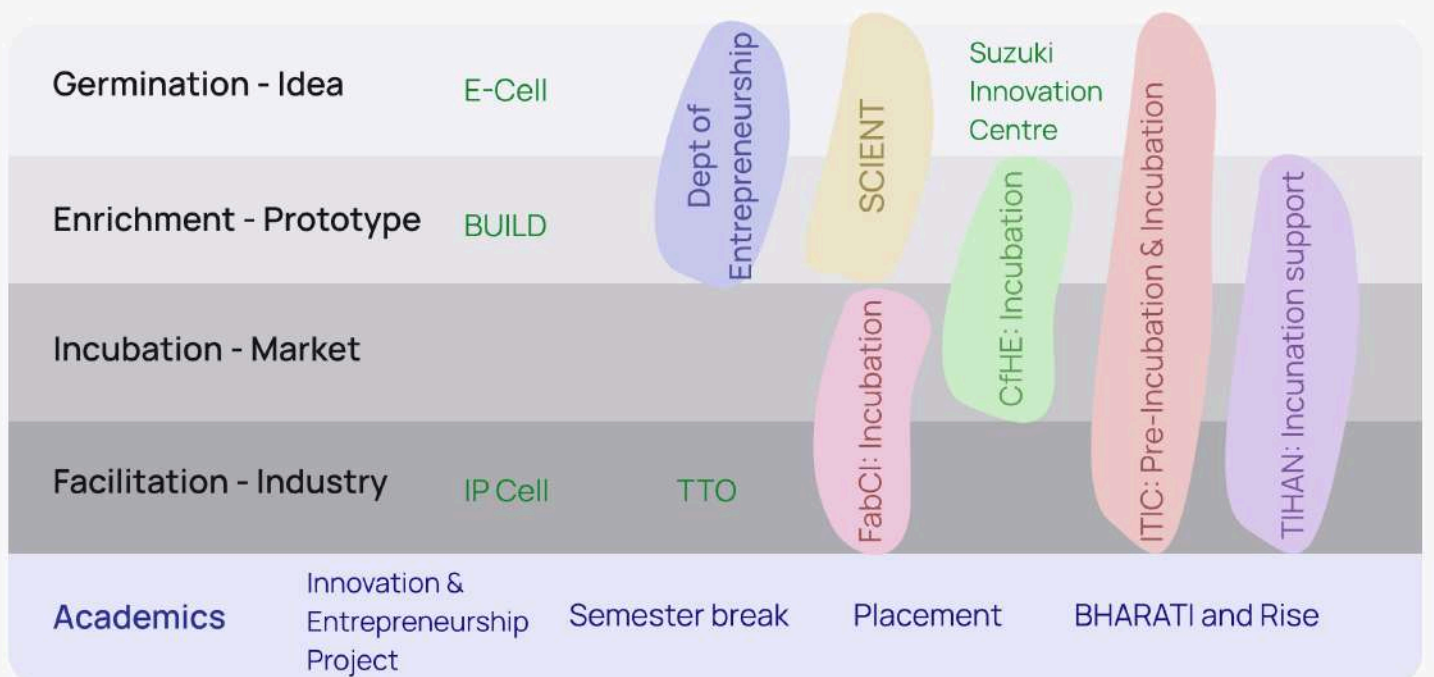
India - Kenya Innovation Nexus (IKIN)



Innovation Support

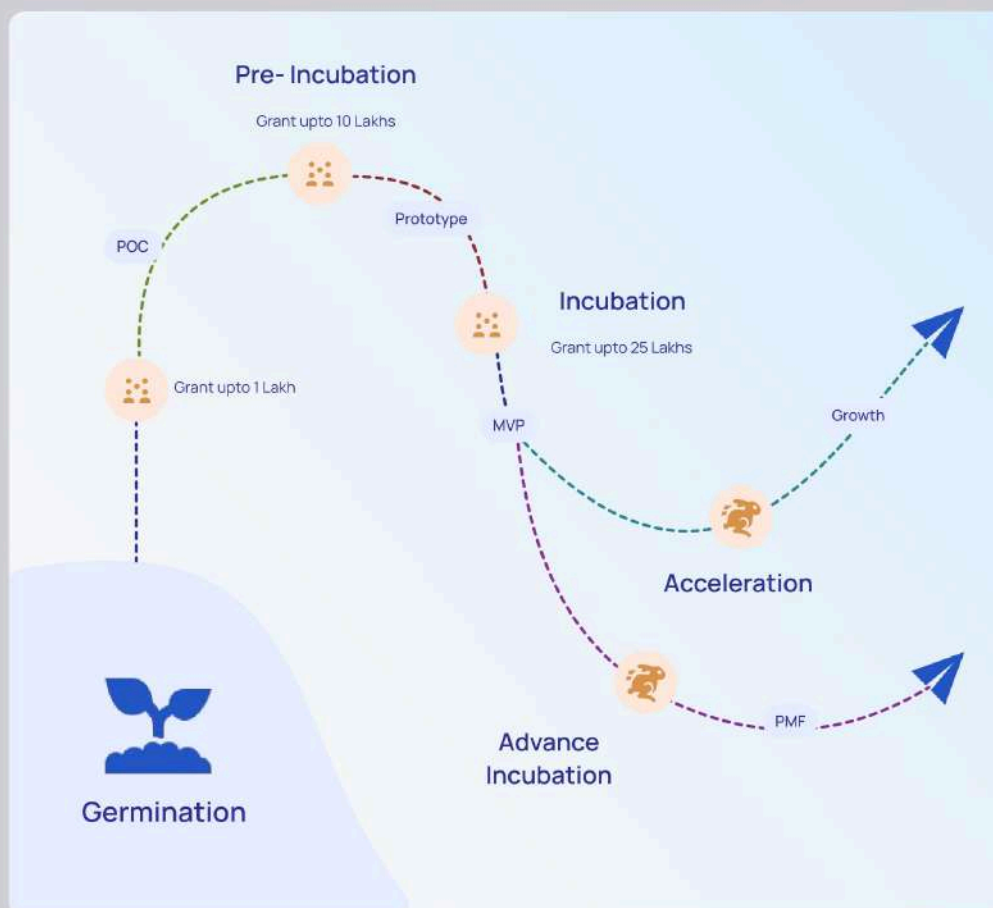


Innovation ecosystem at IIT Hyderabad



The iTIC Incubator at IIT Hyderabad is a DST-funded startup support hub that nurtures deep-tech innovation and entrepreneurship. It provides early-stage founders with incubation support, including seed funding, mentorship, technical guidance, lab access, and industry connections. With a strong focus on cutting-edge technologies and innovation-driven impact, iTIC enables startups to develop, validate, and scale solutions across sectors, strengthening the entrepreneurial ecosystem at IIT Hyderabad and beyond.

Deep-tech | IoT | AI Energy | Healthcare | 5G Mobility





Technology Innovation Hub for Autonomous Navigation

The iTIC Incubator at IIT Hyderabad is a DST-funded startup support hub that nurtures deep-tech innovation and entrepreneurship. It provides early-stage founders with incubation support, including seed funding, mentorship, technical guidance, lab access, and industry connections. With a strong focus on cutting-edge technologies and innovation-driven impact, iTIC enables startups to develop, validate, and scale solutions across sectors, strengthening the entrepreneurial ecosystem at IIT Hyderabad and beyond.



TIHAN activities

Industry collaboration

Innovation

Research & Technology

Entrepreneurship

Navigation testbed

Collaboration

Skill development



The Center for Healthcare Entrepreneurship (CfHE) at IIT Hyderabad is a dedicated hub that nurtures affordable and accessible healthcare innovations. Its one-year full-time fellowship enables innovators to develop market-ready medical technologies through clinical immersion, prototyping, and industry mentorship. CfHE strengthens India's medical innovation ecosystem by transforming healthcare challenges into impactful solutions.



1 Year, In-Resident

HRA/Stipend support

12 fellows, 6 startups

Access to facility

25L seed grants

Clinical & Regulatory

Funding support

IIT Hyderabad Technology Incubation Center (**iTIC**) is a Technology Business Incubator (TBI) established under the aegis of Indian Institute of Technology Hyderabad (IITH).

It operates as a not-for-profit society and was initially supported by the Department of Science and Technology (DST), Government of India.

While primarily focused on deep-tech innovation, its initiatives cover a broad range of emerging and interdisciplinary domains. Through national programs and incubation schemes, iTIC empowers entrepreneurs to transform research-driven ideas into scalable, impactful ventures that contribute to India's innovation ecosystem.

iTIC fosters technology-led entrepreneurship by offering startups - funding opportunities, advanced infrastructure, expert mentorship, and access to extensive industry and investor networks.

205+

Startups Supported

₹ 5+ Crore

Direct Funds to Startups

₹ 50+ Crore

External Funds Raised by Startups

₹ 1200+ Crore

Revenue Generated by Incubated Startups

1000+

Jobs created by Startups

IIT HYDERABAD TECHNOLOGY INCUBATION CENTER



CURRENT FLAGSHIP PROGRAMS AT iTIC

NIDHI PRAYAS - DST, Government of India

Support: Prototype grants up to ₹10 lakh for innovators and early-stage startups to develop working prototypes, including mentoring, lab access, and milestone-based evaluations.

Outcome: Over 40 innovators nurtured; around 30+ evolved into startups, filed patents, or secured further funding.

Vision: Empowering first-time innovators to transform ideas into tangible innovations.



TIDE 2.0 - MeitY (Ministry of Electronics & IT)

Support: Funding up to ₹4 lakh (EiR) and ₹7 lakh (Grant) per startup in IoT, AI, and deep-tech, along with business mentorship and access to IIT Hyderabad's R&D infrastructure.

Outcome: 10+ startups developing commercially viable technologies.

Vision: Driving inclusive digital transformation through innovation.



TiHAN Startup Support Program - NM-ICPS (DST) & IIT Hyderabad

Support: Grants up to ₹10 lakh and funding up to ₹25 lakh for startups in autonomous systems, drones and robotics. Access to test tracks and labs.

Outcome: 13+ startups developed prototypes and real-world smart mobility applications.

Vision: Accelerating India's autonomous and intelligent technology solutions.



Startup India Seed Fund Scheme (SISFS) - DPIIT

Support: Grants up to ₹10 lakh and seed funding up to ₹50 lakh for proof-of-concept, prototype development ensuring market readiness and structured mentorship.

Outcome: Startups scaling innovations across sectors and progressing toward commercialization.

Vision: Fueling India's innovation economy with early capital support.



iDEX - Innovations for Defence Excellence Program - DIO Department of Defence Production, Ministry of Defence

Support: Incubation, Mentorship, infrastructure, and funding from Defence Innovation Organization(DIO).

Outcome: Startups receive funding support, defence-driven product feedback, pilot deployment opportunities, and potential procurement once the solution is fully developed.

Vision: Driving self-reliance in defence technology through collaboration and validation.



FLAGSHIP PROGRAMS SUCCESSFULLY DELIVERED

NICE Program - NMDC & iTIC

Support: Funding through two tracks - NICE Fellowship (up to ₹9.6 lakh) and NICE Incubation Grant (up to ₹25 lakh) along with mentorship, workspace, IITH infrastructure access, and industry connects from NMDC.

Outcome: Enabled 17 innovators to validate ideas, build prototypes, and progress from Fellowship to Incubation, with several startups advancing toward early market deployment.

Vision: Strengthening deep-tech innovation by supporting founders from idea to prototype, creating a pipeline of market-focused, industry-ready startups.



BUILD Program – Greenko Group & iTIC

Support: Grants up to ₹1 lakh for student innovators focusing on sustainability and clean-tech, with mentoring and prototyping support.

Outcome: Over 75 student innovators mentored; many progressing to startup incubation.

Vision: Cultivating future innovators for a sustainable future.



AISEA – AFI iTIC Foundation Social Entrepreneurship Accelerator

Support: Mentorship, investor connects, and strategic partnerships for technology-driven social enterprises in health, edtech, and agritech.

Outcome: 29 startups graduated with measurable social and business impact.

Vision: Accelerating inclusive, socially impactful innovation.



BEL Seed Fund – Bharat Electronics Limited, Ministry of Defence

Support: Seed funding and mentorship for startups developing defence and communication systems, enabling co-development and validation.

Outcome: Startups advancing prototypes, securing patents, and defence contracts.

Vision: Bridging innovation to national defence needs.



ABCD – Acclimatization Boot Camp for Defence Startups

Support: A 12-week program providing expert defence mentorship, user insights from armed forces, and access to IIT-Hyderabad's prototyping ecosystem.

Outcome: Startups refine their prototypes into defence-ready solutions and become fully prepared to apply for iDEX, DISC, and other defence innovation opportunities.

Vision: To transform dual-use technologies into impactful defence innovations that strengthen India's strategic and technological capabilities.



EVENTS AT iTIC

Incubator Development Program (IDP)

The Incubator Development Program (IDP) at IIT Hyderabad, organized by iTIC, brought together experts from academia, industry, and the startup ecosystem to exchange best practices in incubation, entrepreneurship, and innovation. Through collaborative sessions, expert talks, and interactive activities, participants explored vital topics such as governance, mentorship, legal frameworks, intellectual property, and operational models crucial for building and sustaining vibrant incubator ecosystems. The event fostered knowledge sharing, showcased student-driven innovation, and strengthened institutional networks to advance entrepreneurial growth and impact.



Vishwakarma Awards 2024

The Vishwakarma Awards 2024, organized by the iTIC Incubator at IIT Hyderabad in partnership with Maker Bhavan Foundation, celebrated innovative solutions in Green Technology, Smart Mobility, and Water & Sanitation. The nationwide challenge attracted 528 student teams from 276 colleges, with 18 finalists selected to present their deployable prototypes at IIT Hyderabad. The event offered mentorship, financial support, and industry networking, alongside a residential entrepreneurial training program for finalists. Winning teams received cash prizes, certificates, and opportunities for summer internships with partner organizations, highlighting iTIC's commitment to fostering impactful technology innovations for a sustainable future.



Innovation Day 2024

Innovation Day 2024 was a landmark event celebrating the spirit of entrepreneurship and technology at IIT Hyderabad. Drawing more than 2,000 attendees from across the country, the event featured over 45 startup exhibitors, dynamic showcases of groundbreaking technologies, expert talks, interactive panel discussions, pitch sessions, and extensive networking opportunities. A highlight was the MeitY Grand Challenge Finale, conducted in partnership with SINE IIT Bombay, where the nation's top 10 startups competed and presented transformative ideas across digital twin, AI, and simulation technologies. The event set new benchmarks for inclusivity, using cutting-edge assistive technology to ensure full participation for all, including the deaf and hard-of-hearing community. Through this vibrant gathering, iTIC reaffirmed its dedication to building a collaborative, accessible startup ecosystem.



India-Kenya Innovation Nexus (IKIN)

The India-Kenya Innovation Nexus (IKIN), hosted by the iTIC Incubator at IIT Hyderabad, is a strategic collaboration aimed at strengthening innovation and entrepreneurship ties between India and Kenya. Formalized through a partnership charter signed during a high-profile visit by the Kenya National Innovation Agency (KeNIA) and representatives from 11 Kenyan universities, IKIN promotes cross-border knowledge exchange, startup ecosystem development, and capacity-building initiatives. The two-day event included tours of IIT Hyderabad's incubation facilities, discussions on technology commercialization, and academic entrepreneurship, culminating in the joint declaration of membership to the Nexus by Indian and Kenyan institutions. This partnership facilitates collaborative research, startup growth, and mutual market access opportunities, reflecting a shared commitment to fostering technology-driven innovation and economic transformation in both countries.



Indo - German Innovators Connect: Tandem 2025

Tandem 2025 is a flagship Indo-German immersion programme supporting young researchers and early-stage startup founders from both countries. Hosted in Hyderabad and Bengaluru and co-organized by iTIC Incubator at IIT Hyderabad and international partners, the two-week programme brings together 30 participants to work in interdisciplinary teams. Through design thinking workshops, expert mentoring, industry visits, and pitch sessions, participants develop market-ready ideas, gain hands-on access to leading innovation ecosystems, and expand their networks with peers, mentors, and investors across India and Germany. This collaborative initiative by DWIH New Delhi and a consortium of university incubators empowers the next generation of science entrepreneurs to build globally relevant solutions and accelerate their international journey.



FACILITIES & SUPPORT AT iTIC



CO-WORKING SPACE



MENTORSHIP



IP SUPPORT



FINANCIAL AID



NETWORKING EVENTS



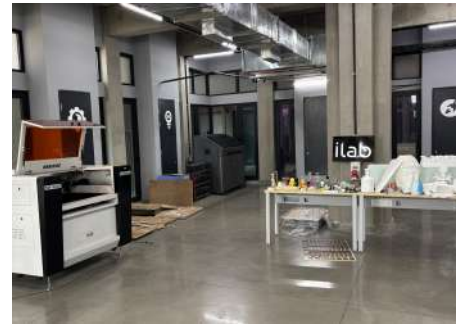
SOFTWARE TOOLS



TECH INFRASTRUCTURE



MAKER LAB



CONNECT WITH US !

✉ office.itic@iith.ac.in

☎ +91 833103 6155

🌐 www.itic.iith.ac.in

📍 iTIC Incubator
Technology Innovation Park, IIT Hyderabad, Kandi,
Sangareddy District, Telangana, India - 502285

📷 [in](#) [f](#) [X](#) /iTICIncubator



PUR Energy Ltd

Founders: Dr. Nishanth Dongari, Rohit Vadera

Sector: **Clean Technology / Electric Mobility**

TRL Level: **TRL 9**

Challenge: India requires reliable, sustainable energy storage to reduce fossil fuel dependency and power outages.

Innovation: PUR Energy's PuREPower Home integrates lithium-ion batteries, inverter, and solar input into a single AI-driven, high-efficiency power system.

Impact: With 320+ dealers nationwide, each system offsets 1.5 tons of CO₂ annually, contributing directly to India's renewable energy and net-zero goals.

Fund Raised

₹140 CR

Turnover

₹137.4 CR

Employment

345

Patents Filed

3



Manali Swing Adventures Pvt Ltd

Founders: Utsav Soni, Siddharth Prasad, Akash Vishwakarma

Sector: **Adventure Tourism / AI-Enabled Safety**

TRL Level: **TRL 6**

Challenge: Most adventure experiences compromise on safety or innovation, limiting India's competitiveness in the global adventure tourism market.

Innovation: Manali Swing introduces an AI-powered, portable high-altitude swing offering over 100 jump styles, combining thrill with real-time safety analytics.

Impact: Pioneers the world's first AI-assisted swing platform, establishing new safety and creativity benchmarks in high-adrenaline tourism.

Fund Raised

₹25 Lakhs

Turnover

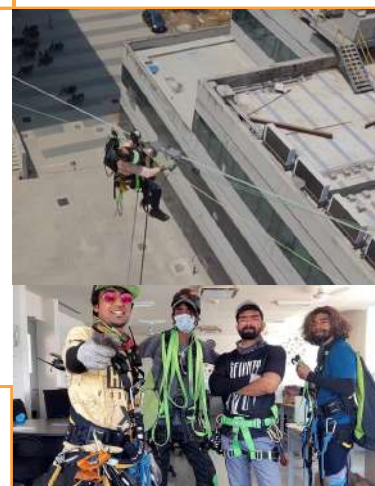
NA

Employment

3

Patents Filed

NA



Brela Innovations Pvt Ltd

Founders: Shreya Nair, Zahid Khan

Sector: **Healthcare / Medical Devices**

TRL Level: **TRL 4**

Challenge: Millions of women skip breast cancer screenings due to cost, discomfort, and lack of privacy.

Innovation: Brela's NaiBra is a wearable, non-ionizing breast screening patch that detects abnormalities in one minute using ML-based 3D imaging.

Impact: Demonstrated 96% accuracy in trials, enabling affordable, private, and early detection for women across diverse communities.

Fund Raised

₹45 Lakhs

Turnover

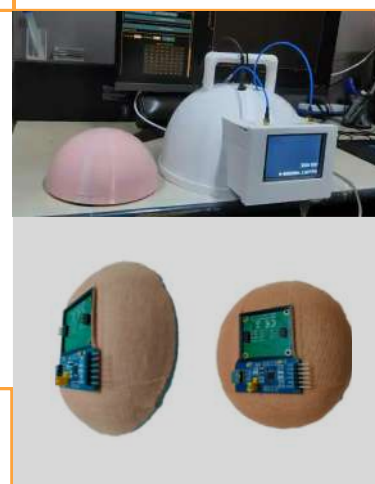
NA

Employment

5

Patents Filed

1



B100M Computer (Bloom)

Founders: Subhajeet Mukherjee

Sector: **Deep Tech, Hardware & Software Systems**

TRL Level: **TRL 7**

Challenge: India's AR and VR industries rely heavily on foreign technologies, leading to limited IP control and dependency on external ecosystems.

Innovation: Bloom is building India's first sovereign computing platform powered by B26 OS - a fully indigenous hardware-software ecosystem with RISC-V-based processors in development.

Impact: Having achieved TRL 7, Bloom is piloting deployments in academia and defence, positioning India toward complete digital self-reliance in computing infrastructure.

Fund Raised

₹10 Lakhs

Turnover

NA

Employment

2

Patents Filed

1



Extrive Innovations Pvt Ltd

Founders: Abhishek, Yashaswi, Prem, Ronak

Sector: **Industrial Ergonomics, Wearable Technology**

TRL Level: **TRL 6**

Challenge: Workers in manufacturing and logistics face chronic back strain and injuries due to heavy manual labor and poor ergonomic tools.

Innovation: Extrive's BackEX is India's lightest exosuit (0.7 kg), reducing lower-back strain by up to 48% and priced affordably at ₹5,499 for mass adoption.

Impact: Deployed across industry leaders such as Maruti Suzuki, Mahindra, and L&T, it redefines industrial safety by improving productivity and worker well-being.

Fund Raised

₹32.2 Lakhs

Turnover

₹10 Lakhs

Employment

9

Patents Filed

1



Pavakah Energy Pvt Ltd

Founders: Vikram Bakaraju, Kapardhi Bharadwaj

Sector: **Cleantech & Advanced Sustainable Materials**

TRL Level: **TRL 7**

Challenge: Traditional solar panels remain rigid, heavy, and costly to deploy, limiting adoption in diverse environments that demand flexible renewable solutions.

Innovation: Pavakah Energy is developing a nanotechnology-based thin film capable of converting any surface into a solar panel while reflecting UV and IR rays to enhance energy efficiency.

Impact: Recognized by Vignan Dhara-PSA, Government of India, and showcased at TechSparks 2025, the innovation has drawn strong industrial interest for advancing energy independence.

Fund Raised

₹2.55 CR

Turnover

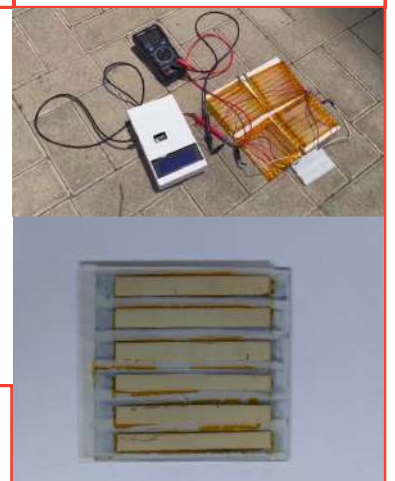
NA

Employment

5

Patents Filed

1



KLVIN Technology Labs Pvt Ltd (KLVIN.ai)

Founders: Vinay Kumar Kolusu

Sector: **Industrial IoT / Predictive Maintenance**

TRL Level: **TRL 6**

Challenge: Industries suffer major losses due to unexpected machinery breakdowns and reactive maintenance models.

Innovation: KLVIN.ai's SENTINEL platform uses patented edge-AI sensor fusion to monitor motors in real time, predicting faults and enabling proactive maintenance.

Impact: The solution enhances energy efficiency, safety, and reliability, with pilots running across steel and rubber industries, targeting over 300 motor deployments within a year.



Fund Raised
₹52 Lakhs

Turnover
NA

Employment
5

Patents Filed
1

LiqSure Systems Pvt Ltd

Founders: Dr. Sarjerao Bapu Doltade, Dr. Archana Doltade

Sector: **Wastewater Treatment/Environmental Technology**

TRL Level: **TRL 9**

Challenge: Industrial wastewater treatment remains costly, chemical-intensive, and inefficient, often creating secondary waste.

Innovation: LiqSure's cavitation-based system treats effluents using only electricity—no chemicals, membranes, or filters—within a compact 10 sq. ft. unit.

Impact: Operational across textile and urban wastewater sites, the zero-chemical process significantly reduces costs and carbon footprint while enabling sustainable water reuse.



Fund Raised
₹30 Lakhs

Turnover
₹1.14 CR

Employment
10

Patents Filed
5

WiSig Networks Pvt Ltd

Founders: Prof. Kiran Kumar Kuchi

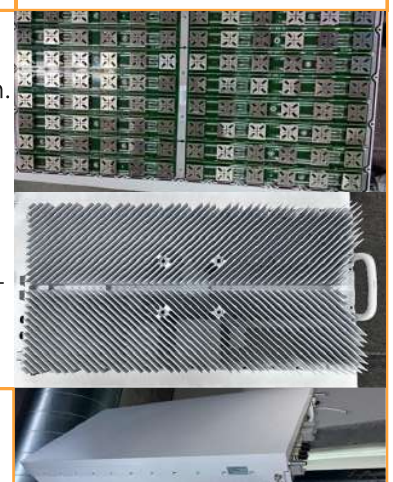
Sector: **Telecommunications / 5G / DeepTech**

TRL Level: **TRL 8**

Challenge: Challenge: India's telecom sector depends heavily on imported 5G infrastructure, driving up costs and limiting local innovation.

Innovation: WiSig Networks has built India's first indigenous 5G/6G IPs, NB-IoT & Satcom SoCs, and carrier-grade 5G RAN equipment, all compliant with global standards like 3GPP and GCF.

Impact: With 250+ patents (including 12 SEPs), India's first compliant NB-IoT & Satcom modem, and 20+ deployed 5G RAN sites, WiSig is enabling true telecom self-reliance.



Fund Raised
₹52.5 CR

Turnover
₹4.05 CR

Employment
100+

Patents Filed
250+



**Center for Healthcare
Entrepreneurship**

"Accessible, Adequate & Affordable Healthcare"



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Center for Healthcare Entrepreneurship, IIT Hyderabad

DST Approved Technology Business Incubator;
DBT-BIRAC BioNEST for Medical Technology;
Medical devices training and Skill Development
Center; Healthcare Entrepreneurship Educator;
Section 8 company with 80G and 12A Registration



Scan to enquire now
Room no 401, 4th Floor,
Foundation for CfHE
IIT Hyderabad, Kandi
Sangareddy, India, 520085



Center for Healthcare
Entrepreneurship

STATE OF THE ART FACILITIES FOR MEDTECH INNOVATION

ADVANCED PROTOTYPING & MANUFACTURING

- 3D Printers (Rapid prototyping for MedTech devices)
- Acrylic Laser Cutting & Vacuum Forming
- Additive & Subtractive Prototyping (for custom device fabrication)
- Metal Tooling & Batch Manufacturing
- CNC Machines (VMC & Lathe) & Metal Laser Cutting

MECHANICAL & FABRICATION TOOLS

- Cutting, Grinding, and Gas Welding
- EDM Gauge Cutter & Hydraulic Machines
- High-End Workstations for Design & Simulation

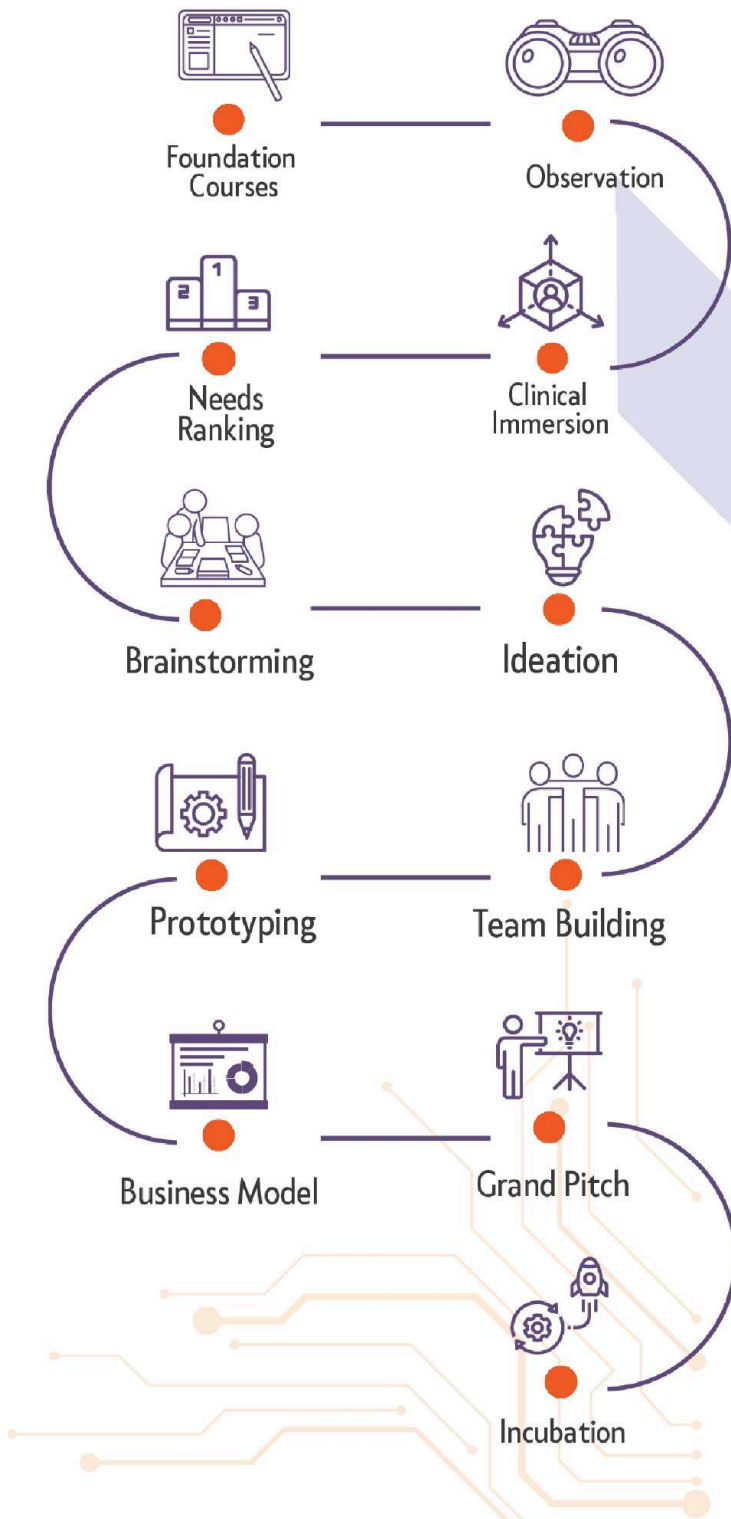
SOFTWARE & DIGITAL DESIGN TOOLS

- Autodesk Manufacturing Suite & Adobe Creative Cloud
- Solidworks (2018/2019/2023) & Proteus Pro
- Development Boards, SBCs, and SOCs for Embedded Systems

ELECTRONICS & TESTING EQUIPMENT

- ESD Workstations & Digital Signal Generators
- Oscilloscopes for Circuit Analysis
- Medical Device Simulators & Pre-Compliance Testing
- Electronics RoHS Compliance Testing

HEALTHCARE INNOVATION AND ENTREPRENEURSHIP FELLOWSHIP



BOARD OF DIRECTORS

Prof. B. S. Murty

Director IIT Hyderabad

Mr. Avi Manudhane Nash

Former Partner, Goldman Sachs (Founder, CfHE)

Mr. Kirat Madhusudhan Patel

Exec. Director, Alkyl Amines Pvt. Ltd.

Mr. Mayur Anand Sirdesai

Co-Founder Somerset Indus Capital Partners

Dr. Nageshwar Reddy Duvvur

Chairman, AIG Hospital

Mr. Prashant Narayan Ranade

Former co-Chairmen Syntel Inc. Co-Founder IndusSME

Prof P. V. Madhusudhan Rao

Head, Department of Design, IIT Delhi

Mr. Rajesh Mashruwala

Prime Venture Partners, Ex VP, TIBCO (Founder, CfHE)

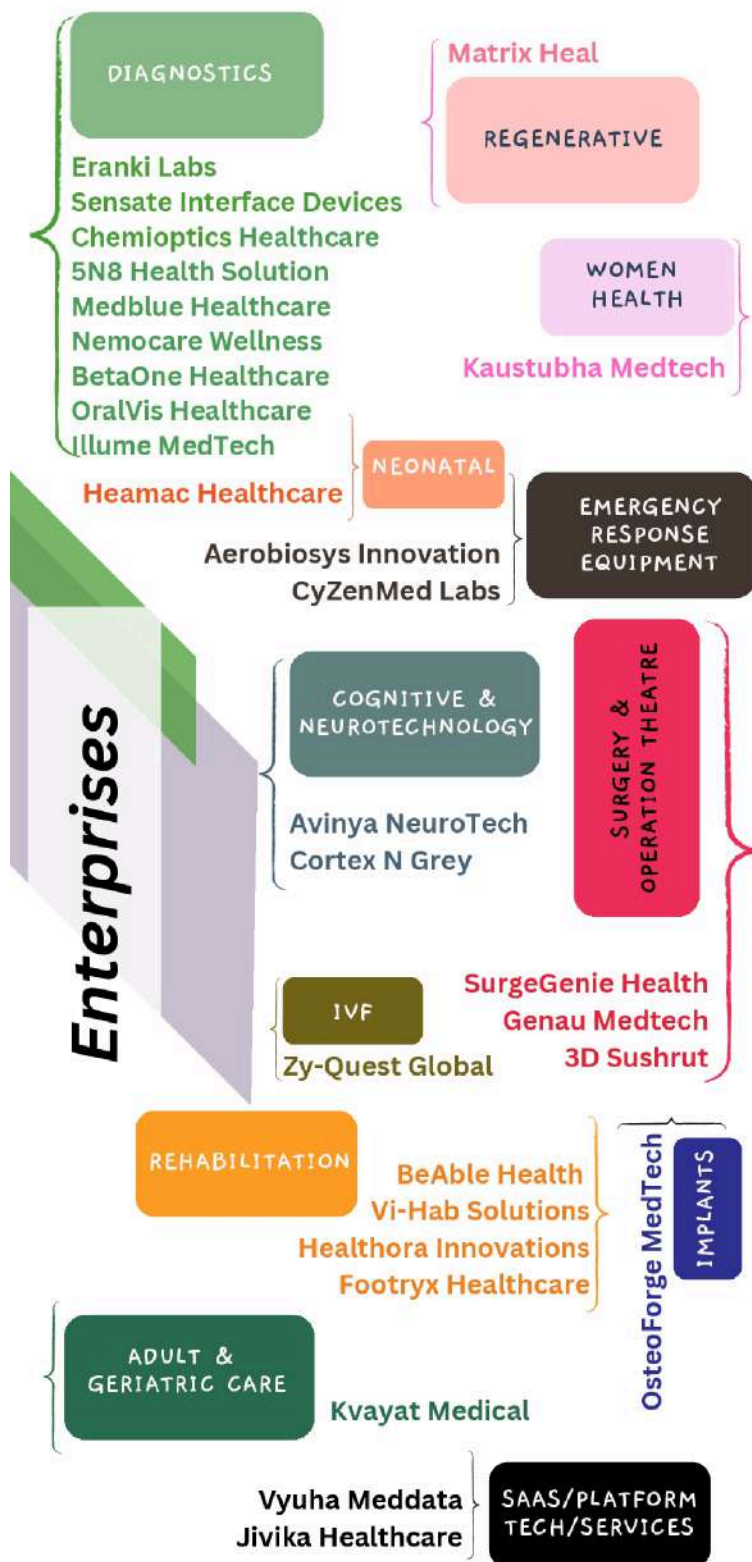
Prof. Sushmee Badhulika

Professor, Dept. of Electrical Engineering, IIT Hyderabad

Prof. Renu John

Professor, IIT Hyderabad (Founder Director, CfHE)





SUCCESS STORIES OF CfHE STARTUPS

Beable Health Pvt Ltd

Founders: Habib Ali, Shreehari Girija

Sector: **Healthcare**

TRL Level: **TRL 9**

Challenge: There is a growing need for intensive, engaging, and high-quality rehabilitation therapy for individuals with upper-extremity motor deficits—both in clinical settings and at home. Current systems place a heavy workload on therapists, creating gaps in therapy intensity and continuity. A solution is required that can deliver effective rehabilitation while reducing the burden on therapists without compromising treatment quality.

Innovation: Game based Upper Limb Rehabilitation Device for Neuroplasticity- Need for an intensive and engaging, rehabilitation therapy for individuals with upper extremity motor deficit at clinic and home, which can minimize the burden on the therapists, without compromising on the quality of therapy.



Fund Raised
₹5 CR

Device Deployed
200+

Employment
15

Patents Filed
NA

Nemocare Wellness Pvt Ltd

Founders: Manoj Sankar, Pratyusha Pareddy

Sector: **Healthcare**

TRL Level: **TRL 9**

Challenge: In India, 99% of differently abled (neurodevelopmental disordered) children, underwent screening and diagnosis at a significantly delayed stage.

Innovation: Neonatal wearable vital parameter monitoring and AI powered decision support : First of its kind IoT enabled medical grade wearable for new-born babies backed by our AI engine that monitors 6 key vital signs, identifies and pre-emptively predicts critical distress conditions and gives the complete picture of the baby's health



Fund Raised
₹7 CR

Device Deployed
100+

Employment
8

Patents Filed
NA

SUCCESS STORIES OF CfHE STARTUPS

Heamac Healthcare Pvt Ltd

Founders: Akitha Kolluju, Prasad Muddam

Sector: **Healthcare**

TRL Level: **TRL 9**

Challenge: 1.5Cr babies in India are born with jaundice among them 60Lakh babies are suffering from severe jaundice where there is no device to treat such cases which are leading to Blood transfusions, brain damage or deaths.

Innovation: nLite-360™ is intelligent enough to provide customized therapy which solves the critical conditions and reduces up to 50% of therapy time while providing uninterrupted treatment while breast feeding and providing kangaroo mother care.



Fund Raised

₹6 CR

Device Deployed

100

Employment

15

Patents Filed

NA

Aerobiosys Innovations Pvt Ltd

Founders: Cyril Antony, Rajesh

Sector: **Healthcare**

TRL Level: **TRL 9**

Challenge: Delivering affordable, reliable, and ICU-grade respiratory-care devices that can function effectively in resource-limited settings, where intermittent power, shortage of trained personnel, high equipment costs, and limited access to quality respiratory support continue to restrict timely and equitable care.

Innovation: Low cost ventilator: A life-saving medical equipment with advanced technologies for hospitals, medical institutions, and individual practitioners.



Fund Raised

₹6 CR

Device Deployed

200

Employment

25

Patents Filed

NA



NMICPS TiHAN FOUNDATION IIT HYDERABAD

| NextGen Autonomous Navigation

Department of Science & Technology (DST), Government of India
National Mission on Interdisciplinary Cyber-Physical Systems (NMICPS)

Technology Innovation Hub on Autonomous Navigation and
Data Acquisition Systems (TiHAN) Foundation



tihan.iith.ac.in

TiHAN-IITH

Vision

Hub for Safe, Sustainable, Smart Next Generation Mobility

Real-time Autonomous Navigation and Data Acquisition Systems
(Unmanned - UAVs, ROVs, etc.)

Quality Data Acquisition for Aerial/Terrestrial Mobile Environments
Multi-sensory Perception Scenario (UAVs, ROVs.), AI Framework,
Real-time Edge Compute Architecture, Communication
Networking, Testing Validation, Simulators (Virtual/Physical), Design

Standard Operating Procedures (Autonomous UAVs, ROVs, etc.)

Testbed/Living Labs for Autonomous Navigation
Systems (Aerial and Terrestrial Vehicles)

Autonomous Transportation
Systems
(Aerial/ Terrestrial/Surface)

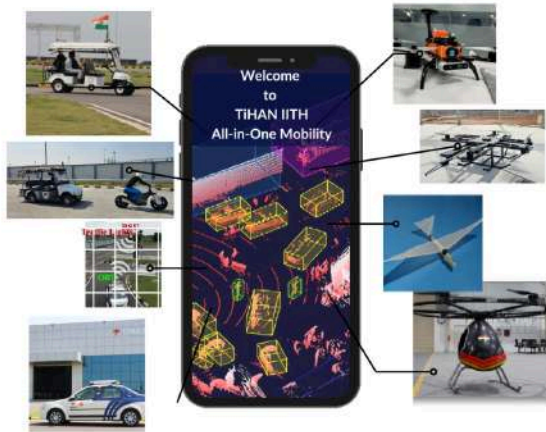
Agriculture & Farming Infrastructure
& Environment & Defence
& Surveillance

Re-imagine ways of working with industry and academic partners to accelerate change

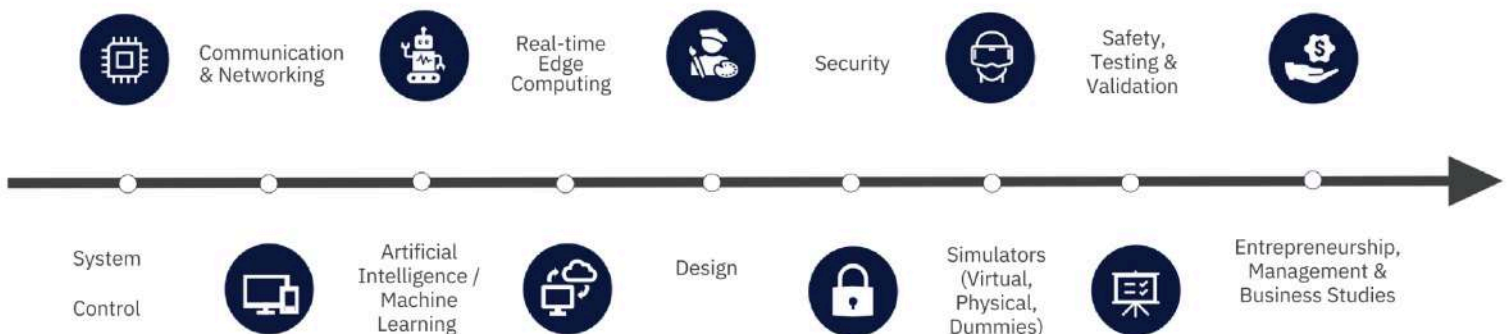
TiHAN-IITH

Activities

TiHAN - IITH: Source for fundamental knowledge and technologies (IPs, Publications, Products, Commercialisation as Licencing, ToTs.....) in the technology vertical of Autonomous Navigation and Data Acquisition Systems.



TiHAN Core Research/Work Groups

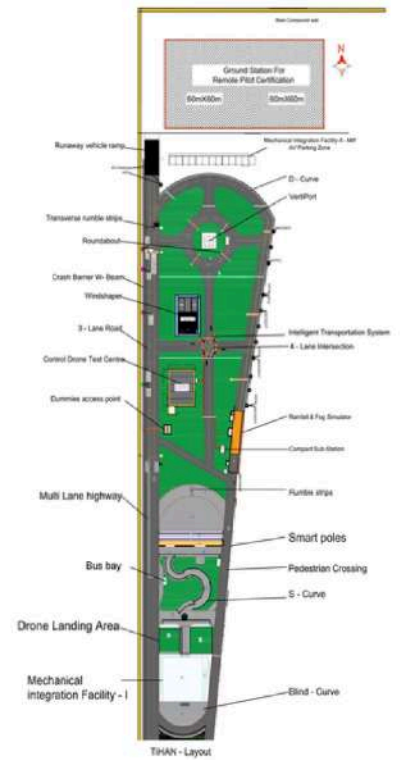


Autonomous Navigation Testbed (Aerial/Terrestrial)

A first of its kind state-of-the-art testbed for Autonomous Navigation (Aerial/Terrestrial)

Technology development and thorough validation before going for real field deployment

Facilities include – Proving Grounds, Test tracks & circuit, Mechanical integration facilities like Hangers, Ground control stations, State of the art Simulation tools (SIL, MIL, HIL, VIL), Road Infra –Smart Poles, signalized & unsignalized Intersections, Environment Emulators like Rainfall Simulators , V2X Communications, Drone Runways & Landing area, Control Test centres, Edge cloud, Multi Radio Access Technologies such as WiFi, BVLOS, CV2X, 5G, DSRC.

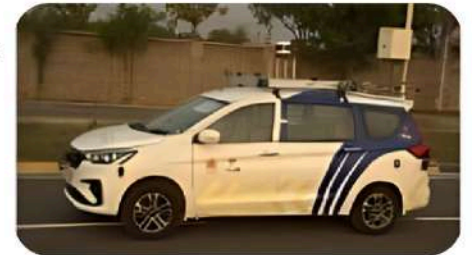


TiAND Dataset

TiHAN has launched the TiHAN-IITH Autonomous Navigation Dataset (TIAND) Dataset for Download. (<https://tihan.iith.ac.in/tiand-datasets/>)

- Data Collection Vehicles: Aerial, Terrestrial and Underwater.
- Datasets offered: Agriculture; Indian roads; Underwater.

The dataset from Hyderabad, India, integrates structured and unstructured traffic data from four cameras, six radars, a Lidar, and GPS/IMU technologies, providing a comprehensive multimodal perspective for enhancing object detection algorithms.



Technology Development

TiHAN at IITH focuses on Unmanned Aerial Vehicles (UAVs) and Unmanned Ground Vehicles (UGVs) for a wide range of applications.

UAVs

In Nano/Micro category drones, Bio-Inspired drones like Quad-wing UAV (Dragonfly based) and Flapping Wing Micro Aerial Vehicles (Aerial Birds based). In Medium/Large category drones, TiHAN is focusing on developing solutions for next generation urban air mobility – air taxis, air metros, air ambulances etc. Harnessing the 5G and edge cloud technologies, these UAVs enable precision landing, swarm coordination, and real-time decision-making, pushing the boundaries of aerial innovation.

UGVs

TiHAN focuses on advancing autonomous technologies using UGVs equipped with LiDAR, cameras, and RADAR sensors for Drive-By-Wire enabled path planning and control. AD features are built into passenger vehicle, campus shuttles, and bicycles (for last - mile connectivity). AD/ADAS features includes GPS based Navigation, Map based Navigation, VRU (pedestrian, bicyclist), Motorcyclist & Car detection, emergency braking, LDWS, LKA. Used test scenario of ADAS function assessments in SIL framework for an Indian setting. Connected Vehicles in the CV2X and 5G Network for various V2V, V2I and V2X algorithms for obstacle avoidance and path planning are being developed .



Testing ISO Compliant

TiHAN advances safe, smart, and sustainable mobility by integrating UNECE standards and Euro NCAP scenarios to test ADAS features like AEB, FCW, and ACC. With Vehicle-in-Loop testing, we ensure safety and reliability, aiming to address India-specific driving challenges and enhance innovation in mobility solutions while prioritizing vulnerable road user protection.



EuroNCAP Certified Soft Targets for Active Safety

Going Forward

End to End Connectivity : Multi-modal Transportation Solutions for Indian Scenarios

Energy efficient autonomous navigation enabled different modes of Electric Vehicles: Last mile connectivity, Shared Shuttle services, Urban Air Mobility – personalized air vehicles



About TiHAN Foundation

- NMICPS Technology Innovation Hub on Autonomous Navigation Foundation is registered to avail CSR funds & we solicit grants from eligible companies inline to their continued patronage for CSR activities
- Contributions to TiHAN Foundation are exempt from income tax under section 80G of Income Tax Act of 1961, (India)
- TiHAN is recognized as a Scientific and Industrial Research Organisation (SIRO) by the Department of Scientific and Industrial Research (DSIR) under the Ministry of Science and Technology, Government of India.
- TiHAN is registered under Foreign Contribution (Regulation) Act 2010 by Ministry of Home Affairs, (FCRA Wing) Govt. of India.

CONTACT US



IIT Hyderabad, Kandi, Sangareddy, Telangana, 502284, India



+91 8331040452, +91 8331040459



office.tihan@iith.ac.in, tihan.heo@iith.ac.in, tihan.pd@iith.ac.in



tihan.iith.ac.in

SUCCESS STORIES OF TIHAN STARTUPS

PredictML.AI

Founders: Prabha Sunder, Sunder Muthevi

Sector: AI Robotics, Education

TRL Level: TRL 6

Challenge: Robotics across sectors—drones, service robots, and humanoids—still depend heavily on manual control and basic automation. Their limited ability to interpret environments, make decisions, or navigate autonomously restricts real-world deployment, especially in complex or dynamic settings like schools, industries, or public spaces. Traditional ecosystems, particularly education, are also slow to adopt advanced robotics, making it difficult for emerging solutions to gain trust and demonstrate value.

Innovation: PredictML.AI addresses this gap through AMVIN, an advanced Autonomous Multi-Model Vision AI Navigator that brings real-time 3D vision, obstacle avoidance, and fully offline decision-making to robots. This technology enables machines to perceive, understand, and act independently across land, air, and sea. Beyond navigation, the startup is advancing humanoid intelligence—robots that can interact naturally with people, perform administrative and operational tasks, and continuously learn through contextual awareness.

Vision: The technology is already transforming educational environments by deploying AI-powered humanoids that guide visitors, support admission processes, assist staff, and enhance student engagement. Early partnerships with institutions such as IIT Bhubaneswar and progressive schools validate both the capability and scalability of the solution. With strong traction and a clear roadmap, PredictML.AI is positioned to bring intelligent robots into mainstream operations across education, hospitality, and healthcare.



SUCCESS STORIES OF TİHAN STARTUPS

Visron Pvt Ltd

Founders: Vinay Yadav, Hitesh Borana

Sector: **Drones, Agritech**

TRL Level: **TRL 6**

Challenge: India's agriculture and drone ecosystems face significant limitations due to short flight endurance, low payload capacity, and high operational costs associated with conventional battery-powered drones. Farmers, especially in remote or high-demand regions, struggle to access reliable aerial spraying and seeding solutions that can perform efficiently across large fields and challenging terrains. Simultaneously, the country remains heavily dependent on imported energy systems for advanced UAVs, creating gaps in self-reliance and scalability for long-duration drone operations.

Innovation: Visron Private Limited is advancing next-generation drone and energy systems through deep in-house R&D and indigenous technology development. The company has engineered its own lightweight, high-strength drone structures using modern manufacturing processes, enabling superior durability and greater payload capacity. These advancements also complement their earlier achievement of deploying Karnataka's first 30 kg payload agriculture drone specifically built for high-volume pesticide spraying and precision farming.

Vision: Visron's innovations are directly transforming agricultural operations and advancing India's clean-tech capabilities. Their high-payload drones enable faster field coverage, reduced chemical wastage, and improved farmer safety by minimizing human exposure to pesticides. By building critical components domestically and focusing on scalable, high-performance UAV platforms, Visron is strengthening India's position in sustainable energy, agricultural automation, and next-generation drone technology.

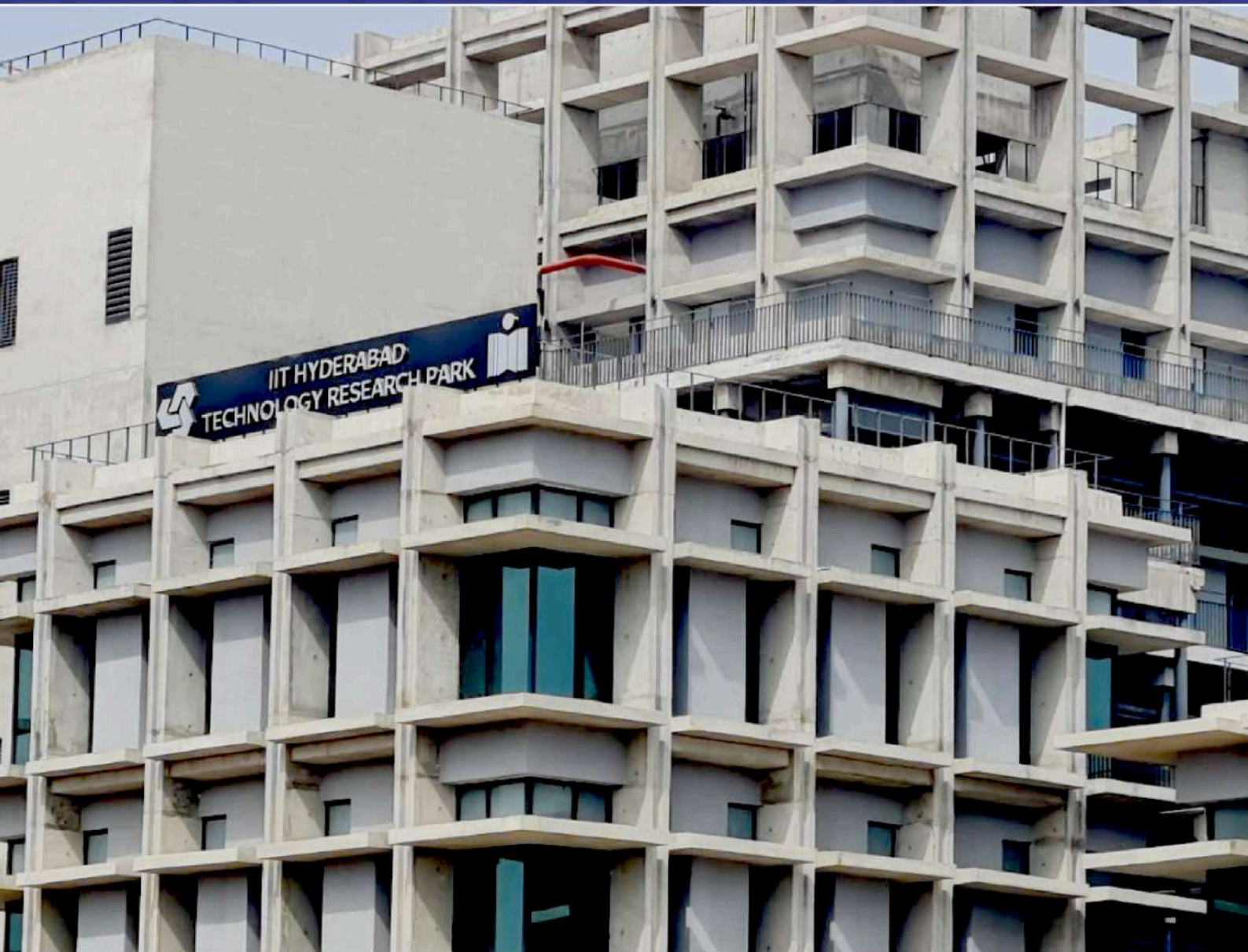




भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

TRP

IIT HYDERABAD TECHNOLOGY RESEARCH PARK



December 2022

About Us

IIT Hyderabad is one of the premier academic institutions in India with a motto of **Inventing and Innovating in Technology for Humanity (IITH)**. **IITH Technology Research Park (TRP)**, situated in the IITH campus aims to provide infrastructure and facilities for industry partners to co-locate Research and Development (R&D) centers at the IITH campus. Funded by the Ministry of Education, Govt. of India (erstwhile Ministry of Human Resource Development) to tune of **Rs. 75 Crores** towards capital expenses, this **1.5 Lakhs square feet** building features state-of-the-art facilities on par with global standards.

IITH Technology Research Park (TRP) is administered as an independent Section 8 Company, founded, promoted and hosted by IITH. It is governed by a board of distinguished academicians, faculty of IITH and industry professionals, to facilitate industry interaction with IITH so that innovative technologies can be developed through collaboration.

The mission of **TRP** is :

- » To bridge the gap between industry & academia.
- » To establish a world class innovation hub through industry-IITH collaboration.
- » To provide a strong and robust platform to foster innovations and inventions through collaboration.

Focus Areas and Research Strengths

Additive Manufacturing	Health Care
Artificial Intelligence	Autonomous Navigation
Bioinspired Processes and System	Integrated Computational Engineering
Climate Change	Sensors and Devices
Energy	Smart Cities
Environment & Development	Sustainable Development
Nano-X	Robotics

Ecosystem of IIT Hyderabad

Vibrant Campus

- 600+ Acres of Campus
- 290+ Faculty
- 18 Departments
- 200 State of the art laboratories

Active Research Culture

- Total Project Funding ~ Rs 700 Crore till date
- 300+ongoing projects (1220 project completed)
- 170+ patents
- 8100 Research Publications (1,20,000+ citations)

Host of Multiple Research Clusters and Centres

- Center for Healthcare Entrepreneurship (CfHE)
- Center of Excellence in Sustainable Development (FAST Center of MHRD).
- Center of Interdisciplinary Programmes (CIP)
- Centre of Cyber Physical Systems (CPS)
- Design Innovation Center (DIC)
- DST NM-ICPS Technology Innovation Hub on Autonomous Navigation (TiHAN)
- Fabless Chip Design Incubator (FabCI)
- iTIC Incubator at IIT Hyderabad
- IPR Cell/ IP Facilitation Center
- Nano-X Fabrication facility
- Rural Development Center (RDC)
- Teaching and Learning Center (TLC)
- DRDO DIA Center
- CoE on Transportation Infrastructure
- AI CoE (Kriya)
- HPC Centre (High Performance Computing Data Center)
- CoE on Additive Manufacturing

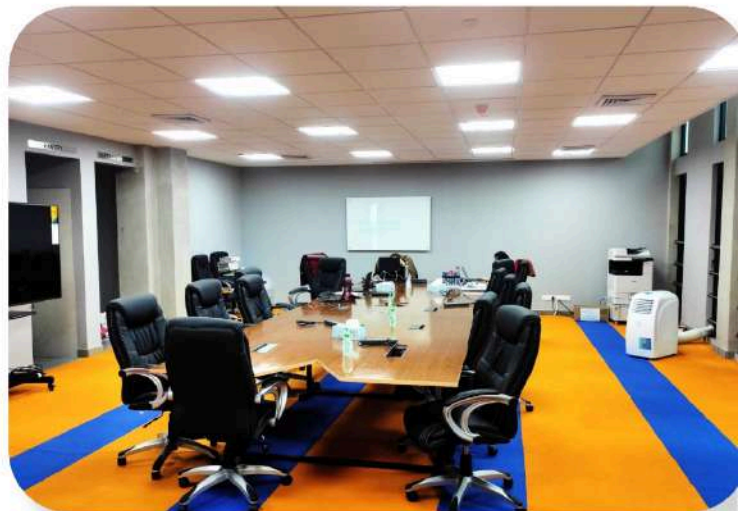
Strong Collaboration with Japan

- Research and Development in identified areas
- Construction support for certain buildings
- Purchase of identified high-end equipment
- Human resource exchange.

For further detailed information please refer IITH website

Occupancy Status

- Total space for labs 55,000 sq ft.
- Available space 30,000 sq ft. of different configurations.



FOR MORE DETAILS:

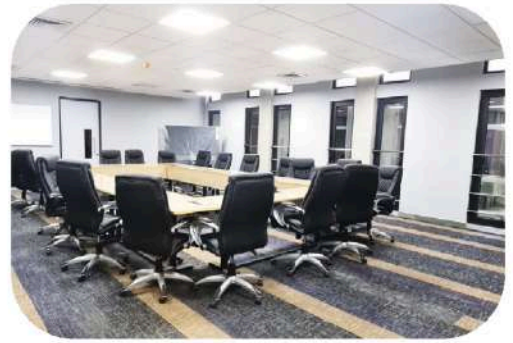
Prof. Sumohana S Channappayya

Faculty-in-charge, Technology Research Park, IIT Hyderabad | **Phone:** 8331036131

Email: fic.trp@iith.ac.in / office.trp@iith.ac.in | **Website:** trp.iith.ac.in

Building & Facilities

- Approximate built-up area is 1,50,000 sft
- 11 individual towers of different levels, connected with corridors.
- Laboratory complex comprising of 7 different sized square footprint blocks interconnected with covered corridors.
- Provides re-configurable spaces & provisions for including wet and dry labs.
- Entrance & courtyard canopies in combination of glass & steel.
- Floors with cantilevered slabs spanning from 0.6m to 1.5m
- Internal courtyard with tensile fabric structure
- Combination of exposed concrete finishes & painted plaster surfaces.
- Dry Research Labs (16 No.)
- Wet Research Labs (16 No.)
- Seminar Rooms (2 No)
- Conference Rooms (15 No)
- Admin Offices (2 Rooms)



Commercials

- Security Deposit (Refundable) - **Rs.1.00 Lakh** per Company (or) three month rental advance (which ever is higher).
- License fee for premises allotted - **Rs.45 per sft.** per month (inclusive of GST)
- Maintenance - **Rs.5900/-** per month (inclusive of GST) for every 1000 sft the space allotted (or) on actuals for large clusters (which ever is higher)
- The Credit Point System is developed to encourage strong interaction between organizations and IITH ecosystem.
- Credit Point System in place for availing benefits (rebate in License Fee).
- Discounted rates for usage of facilities of IIT Hyderabad (Lab, Library, etc.).
- Credit Points maybe earned annually through research collaborations/engagements with IITH ecosystem.
- Credit points earned would be considered in either in the termination/continuance at IITH Technology Research Park.
- An Organization must “earn” a minimum of 200 credit points per year per 1000 sft of space occupied.

Credit Point System

Category	Credits
Sponsored projects to IITH faculty(<i>credit of Rs. 10 lakhs</i>)	200
Royalty (<i>credit of Rs. 1 lakh</i>)	100
Chair Professor	200 per Year
Sponsored MTech/PhD students of IITH	75 per Year
Sponsoring their employees for Masters/PhD at IITH	75 per Year per Student
Campus recruitment of IITH Student	50 per Year per Student
Internship opportunity to IITH Student	25 per Year per Student

Connect with Us



Email

dean.innovation@iith.ac.in
office.innovation@iith.ac.in



Phone

+91 83310 36052



Address

Admin Block – Room No. 413
Chemistry Block – Room No. 417
Indian Institute of Technology Hyderabad, Kandi,
Sangareddy, Telangana 502285



Faculty Development
Program



Incubator Development
Program



Innovation Day



Smart India Hackathon

Event Highlights